

Homework 2 — Written Problems

Quantitative Reasoning II (INF 1220)

Due Wednesday, October 7, 11:59pm, on Populi

Complete these problems by hand. Scan and turn in your work to Populi by the due date.

1. Heat and learning

The tables below are from *Heat and Learning*, a paper by Joshua Goodman, Michael Hurwitz, Jisung Park, and Jonathan Smith (full citation at the end). The authors asked whether hot weather during the school year affects how much students learn. They used PSAT score data from students in the high school classes of 2001–2014 who took the PSAT more than once. Each score is matched to temperatures at the weather station nearest the student’s high school.

Note: There are aspects of this table which we have not covered in class. That’s okay! You can answer the questions without understanding every detail of the table. This is practice for consuming real published research.

- *Outcome*: a student’s PSAT score, math plus reading, *standardized by test administration*. Each score is converted to standard deviations from the average of everyone who took that same PSAT sitting: 0 is exactly average, +1 is one standard deviation above average, -0.5 is half a standard deviation below. The table reports this in *hundredths* of a standard deviation, so a coefficient of -1.00 would mean a score one hundredth of a standard deviation lower (0.01 standard deviations).
- *Mean temperature*: the average daily high, in $^{\circ}\text{F}$, over the school days in the 365 days before the test.
- *Days above 100°F , days in the 90s, . . .*: counts of school days in that year whose high fell in each range.
- *Take number*: students can take the PSAT in 9th, 10th, and 11th grade. In this sample, 85% took it twice and 15% three times.

The *Notes* paragraph under the table is reproduced as published. Reading a paper’s table notes is how you find out what a table contains.

The paper’s abstract includes this sentence: “Without air-conditioning, a 1°F hotter school year reduces that year’s learning by one percent.”

Table 2: Prior Year Temperature and PSAT Scores

	(1)	(2)	(3)	(4)	(5)	(6)
<i>(A) Average heat</i>						
Mean temperature (°F)	−0.181*** (0.028)	−0.211*** (0.036)	−0.185*** (0.029)	−0.182*** (0.028)	−0.176*** (0.027)	−0.230*** (0.042)
<i>(B) Hot days</i>						
Days above 100°F	−0.067*** (0.018)	−0.077*** (0.020)	−0.075*** (0.017)	−0.065*** (0.018)	−0.064*** (0.016)	−0.098*** (0.033)
Days in 90s (°F)	−0.053*** (0.013)	−0.061*** (0.014)	−0.059*** (0.013)	−0.053*** (0.013)	−0.053*** (0.012)	−0.064*** (0.018)
Days in 80s (°F)	−0.035*** (0.009)	−0.037*** (0.010)	−0.039*** (0.009)	−0.035*** (0.009)	−0.029*** (0.009)	−0.046*** (0.013)
Days in 70s (°F)	−0.024*** (0.008)	−0.024*** (0.008)	−0.026*** (0.008)	−0.024*** (0.008)	−0.023*** (0.008)	−0.020* (0.012)
Days below 60°F	0.010 (0.009)	0.013 (0.009)	0.008 (0.008)	0.010 (0.009)	0.010 (0.007)	0.007 (0.011)
<i>N</i>	21,046,448	21,046,448	21,046,448	21,046,448	21,046,448	5,378,273
Prior year snow, rain	No	Yes	No	No	No	No
Test day weather	No	Yes	No	No	No	No
Pollution	No	No	Yes	No	No	No
Economic conditions	No	No	No	Yes	No	No
State-specific trends	No	No	No	No	Yes	No
Sensor within 5 miles	No	No	No	No	No	Yes

Notes: Heteroskedasticity robust standard errors clustered by weather sensor are in parentheses (* $p < .10$ ** $p < .05$ *** $p < .01$). Coefficients in each column and panel come from a regression of hundredths of a standard deviation in PSAT total (math plus reading) scores on the weather measure(s) shown. Temperature is measured with the daily maximum temperature from school days in the 365 days preceding the PSAT take. All regressions include student fixed effects and fixed effects for each combination of cohort, test date and take number. Column 2 adds controls for prior year rainfall and snowfall, as well as test day temperature, rainfall and snowfall. Column 3 controls for prior year and test day pollution levels (carbon monoxide, ozone, sulfur [sic] dioxide, nitrogen dioxide and PM10). Column 4 controls for the logarithm of per capita county-level payroll in industries highly exposed to weather. Column 5 adds state-specific linear time trends. Column 6 limits the sample to high schools within 5 miles of the nearest weather sensor. The sample comprises all students from the high school classes of 2001-14 who took the PSAT more than once.

Table and notes from the working paper version (NBER Working Paper 24639, revised November 2019), reproduced as printed. © 2018 by Joshua Goodman, Michael Hurwitz, Jisung Park, and Jonathan Smith. *Cohort* means high school class.

Rows from Table 1 of the same working paper, which describes the students. Values are means. *All retakers* are the students in Table 2. Table 2's *N* counts test scores, not students: 9.8 million retakers took the PSAT two or three times each, for about 21 million scores.

	All takers	All retakers
Mother has B.A.	0.22	0.33
ZIP code mean income	63.2	69.6
Retook PSAT	0.36	1.00
First PSAT z-score	-0.00	0.14
Mean temperature (°F)	65.1	65.8
Days above 90°F	11.9	12.2
N (scores)	38,303,474	21,076,009
N (students)	27,023,119	9,795,654

ZIP code mean income is mean income in the student's ZIP code as reported by the IRS; the table does not state its units.

1. Interpret -0.181 in column (1). Give it in hundredths of a standard deviation, then in standard deviations.
2. Student fixed effects compare each student's score with that same student's score on another take. Name one thing about a student that this holds constant, and one thing that could be different between a student's two takes.
3. Columns (2) through (5) each add a control. Choose one and say what the authors were worried about when they added it. The Mean temperature coefficient barely changes. What does that tell you about that worry?
4. Build an approximate 95% interval for -0.181 . Is the effect statistically significant at the 5% level?
5. Table 1 shows that retakers' first PSAT scores average 0.14 standard deviations higher than all takers'. Using your answer to question 1, how many degrees hotter would a school year have to be for heat to produce a difference that large? Is the effect of one degree large for a single student? Explain how an effect this small can be statistically significant.
6. Who is in this sample? Use Table 1 to say how they differ from all PSAT takers, and what that means for the students the results describe.
7. Let's say a news site runs this headline:

Every degree of warming costs American students 1% of a year's learning.

Do you think this is an accurate headline? Why or why not?

2. Southwest and arrival delays

This problem uses the flights from Homework 1, departures from Austin between July 2025 and June 2026. It covers the 87,938 flights with an arrival delay recorded, of which 37,924 were Southwest.

	Southwest	Every other airline
Arrival delay, mean (minutes)	8.9	10.4
Arrival delay, median (minutes)	−2	−7
Arrived 15 minutes or more late	26.3%	21.2%
Arrived 3 hours or more late	0.7%	2.3%
Scheduled to depart 5pm or later	29.7%	21.8%

Southwest flies to 58 of the 85 destinations covered in this data. It does not fly to Dallas–Fort Worth, where flights from Austin arrived 23.4 minutes late on average.

Arrival delay is the number of minutes after the scheduled arrival time. A negative value means the flight arrived early: the median of −7 above means the typical flight on another airline arrived 7 minutes early.

Every column below regresses arrival delay in minutes on *Southwest* (1 if the flight is Southwest, 0 otherwise) and whatever else is listed. That is, the dependent variable is arrival delay in minutes. Standard errors are in parentheses (* $p < .10$ ** $p < .05$ *** $p < .01$).

	(1)	(2)	(3)	(4)	(5)
Southwest	−1.47*** (0.43)	−3.42*** (0.44)	−3.46*** (0.44)	−0.43 (0.65)	−1.36*** (0.09)
Departure delay, minutes					1.005*** (0.001)
Departure-hour FE	No	Yes	Yes	Yes	No
Month FE	No	No	Yes	Yes	No
Destination FE	No	No	No	Yes	No
R^2	0.000	0.017	0.023	0.033	0.958
n	87,938	87,938	87,938	87,938	87,938

1. Interpret −1.47 in column (1), with an approximate 95% interval. In Homework 1 you found that Southwest arrived late more often than the other airlines. Use the first table to explain how both are true.
2. What do the departure-hour fixed effects hold constant? What do the destination fixed effects hold constant? Why does the Southwest coefficient grow from (1) to (2) and shrink from (3) to (4)? (*Hint: The first table and the DFW fact may be helpful*).
3. Column (5) has by far the highest R^2 . Why is it the wrong column for the question “if I book Southwest, will I arrive later?”
4. A traveler is choosing between a Southwest flight and another airline’s flight to the same city, leaving at the same hour. Which column answers that? Write the answer as one sentence with an interval.

Sources.

Problem 1. Joshua Goodman, Michael Hurwitz, Jisung Park, and Jonathan Smith, “Heat and Learning,” NBER Working Paper 24639, National Bureau of Economic Research, May 2018, revised November 2019. Tables 1 and 2.

A later version was published as R. Jisung Park, Joshua Goodman, Michael Hurwitz, and Jonathan Smith, “Heat and Learning,” *American Economic Journal: Economic Policy* 12(2), 2020, 306–339. The tables and figures in this problem are from the working paper version.

Problem 2. U.S. Bureau of Transportation Statistics, Reporting Carrier On-Time Performance, July 2025 through June 2026, the Homework 1 file. Regressions computed for this homework.